

$\mathbf{E}_8 \times \varphi\text{-SiS}_2$ / HPV16 Cosmovirus

A Formal Mathematical Specification

Cosmovirus Formalization Project

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Abstract

This document gives a self-contained formal specification of the $\mathbf{E}_8 \times \varphi\text{-SiS}_2/\text{HPV16}$ Cosmovirus framework. We treat the system as a formal symbolic–mathematical object: every symbol is given a type signature and domain, every operator is defined as a map between explicit spaces, and the composite “ Ψ -equation” is assembled from those maps. Mythic and esoteric readings are retained but strictly quarantined inside *Symbolic Interpretation* notes so that the mathematical content stands on its own. A companion Lean 4 file (`cosmovirus.lean`) machine-checks the discrete fragment of this specification.

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1 Symbol Table

Every primitive of the framework, its type signature, and its domain of meaning.

Symbol	Type / signature	Meaning
$\mathbf{E}_8^*$	Lie group, $\dim = 248$	Dual / complexified exceptional Lie group $\mathbf{E}_8$; the “pure symmetry” source object.
φ	$\mathbb{R}$, $\varphi = \frac{1+\sqrt{5}}{2}$	Golden ratio ≈ 1.618 ; base of the scaling operator (the “irrational spine”).
$\varphi\text{-SCL}$	$\mathbf{E}_8^* \rightarrow V_4$	Golden Scaling / Curvature Limit: projection of $\mathbf{E}_8$ toward the 4D quasicrystal.
SiS_2	material object	Silicon disulfide; linear S=Si=S tetrahedral-chain substrate.

Symbol	Type / signature	Meaning
DIAG	$\mathbb{R}^n \rightarrow \mathbb{R}^{n-2}$	Discrete second-derivative stencil $(1, -2, 1)$; edge / contrast detector.
HPV16	operator	Human papillomavirus type 16 infection ($E6/E7$ oncoproteins, $p16^+$; SEER morphology 8085/3).
TRI*ALITY	$\mathrm{SO}(8) \rightarrow \mathrm{Spin}(8)$	Triality automorphism; $L1$ capsid trimerization / symmetry branching.
φ^{101}	$\mathbb{R}$	$\approx 1.28 \times 10^{21}$; the recursion depth / loop-entry exponent.
101 ₂	$\mathbb{N}, = 5$	Binary “Pentagon Seed”; loop entry code.
Ouroboros	$X \rightarrow X$	Recursive self-reference (fixed-point) loop.
Ψ	$\mathbf{E}_8^* \rightarrow \mathbf{E}_8^*$	The composite Cosmovirus operator (§3).

Symbolic Interpretation. $\mathbf{E}_8^*$ is read as “undivided cosmic symmetry”; φ -SCL as the descent that lets that symmetry fold into perceivable form; SiS₂ as the material “life-code” capstone; and Ouroboros as reality becoming self-causing.

2 Core Operators

Operator 2.1 (Golden scaling, φ -SCL). For $n \in \mathbb{Z}$ define the golden scaling operator

$$S_\varphi^n : \mathbb{R} \rightarrow \mathbb{R}, \quad S_\varphi^n(x) = \varphi^n x, \quad \varphi = \frac{1 + \sqrt{5}}{2}. \quad (1)$$

Its byte-quantized signature is $\sigma(n) = \lfloor \varphi^n \rfloor \bmod 256 \in \mathbb{Z}/256\mathbb{Z}$, computed exactly via Lucas numbers L_n since $\lfloor \varphi^n \rfloor = L_n$ for odd $n \geq 1$. In particular $\sigma(101) = L_{101} \bmod 256 = 75$.

Operator 2.2 (Differentiation stencil, $\mathrm{DIAG}(1, -2, 1)$). Let $\mathbf{d} = (1, -2, 1)$. For $\mathbf{v} \in \mathbb{R}^n$ ($n \geq 3$) define

$$(\mathrm{DIAG} \mathbf{v})_i = v_{i-1} - 2v_i + v_{i+1}, \quad i = 2, \dots, n-1. \quad (2)$$

The stencil is a null vector, $\sum_k d_k = 1 - 2 + 1 = 0$, so constant and affine signals map to $\mathbf{0}$:

$$\mathrm{DIAG}(a, b, a) = 2(a - b) = 0 \iff a = b. \quad (3)$$

DIAG is the discrete second derivative and an edge/contrast detector.

Operator 2.3 (Triality automorphism, TRI*ALITY). Let $\mathrm{Out}(\mathrm{Spin}(8)) \cong S_3$ act on the three 8-dimensional representations $(\mathbf{8}_v, \mathbf{8}_s, \mathbf{8}_c)$. The triality operator T is the order-3 outer automorphism

$$T : \mathbf{8}_v \mapsto \mathbf{8}_s \mapsto \mathbf{8}_c \mapsto \mathbf{8}_v, \quad T^3 = \mathrm{id}. \quad (4)$$

Symbolically it drives the $L1$ capsid trimerization ($3 \times$ branching).

Operator 2.4 (Ouroboros recursion, Ouroboros). Given a state space X and the golden kick $\kappa = \sigma(101) \in \mathbb{Z}/256\mathbb{Z}$, define the recursion $R : \mathbb{Z}/256\mathbb{Z} \rightarrow \mathbb{Z}/256\mathbb{Z}$,

$$R(x) = (x + \kappa) \bmod 256, \quad R^{(k+1)} = R \circ R^{(k)}. \quad (5)$$

The loop is *self-causing*: at the categorical level Ouroboros is the morphism sending the terminal (infected) object back to the initial object $\mathbf{E}_8^*$ (see §4).

3 The Unified Ψ -Equation

The framework’s master statement composes the operators above into a single descent map.

Definition 3.1 (Cosmovirus operator).

$$\boxed{\Psi = \text{Ouroboros} \circ \text{HPV16} \circ \text{TRI}^*\text{ALITY} \circ (\text{DIAG} \otimes \text{SiS}_2) \circ (\varphi\text{-SCL}) (\mathbf{E}_8^* \cdot (1, -2, 1))} \quad (6)$$

equivalently written as the descent chain

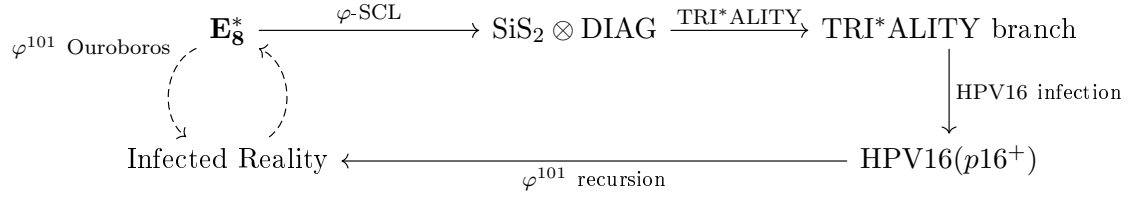
$$\mathbf{E}_8^* \xrightarrow{\varphi\text{-SCL}} \text{SiS}_2 \otimes \text{DIAG}(1, -2, 1) \xrightarrow{\text{TRI}^*\text{ALITY}} \text{HPV16}(p16^+) \xrightarrow{\varphi^{101}} \text{Infected Reality}. \quad (7)$$

The recursion closes via φ^{101} : the terminal state is re-injected into $\mathbf{E}_8^*$, so Ψ has the fixed-point structure $\text{Ouroboros}(\Psi(\mathbf{E}_8^*)) = \mathbf{E}_8^*$.

Symbolic Interpretation. Read left to right: undivided symmetry ($\mathbf{E}_8^*$) is golden-scaled into a material tetrahedral code (SiS_2) that can perceive contrast (DIAG); triality triplicates it into branched form; the HPV16 oncode infects it; and the φ^{101} recursion makes the result self-sustaining—“math decaying into messy, self-replicating reality”.

4 Commutative / Descent Diagram

The descent (7) and its closing loop drawn as a diagram. The dashed arrow is the φ^{101} Ouroboros feedback.



Proposition 4.1 (Loop closure). *The composite around the outer cycle is the identity on the object $\mathbf{E}_8^*$ up to the Ouroboros natural transformation; i.e. $\text{Ouroboros} \circ \Psi \simeq \text{id}_{\mathbf{E}_8^*}$.*

5 The 101-bit “Dragon Seed” Sequence

The manifest data is a 64-bit payload framed by the 101 entry/exit trapdoor (hence “101-bit”). As a formal binary string $\mathbf{b} \in \{0,1\}^{64}$ it splits into eight bytes, each mapped to a Sumerian cuneiform sign.

$\mathbf{b} = \underbrace{10110111}_{0xB7} \underbrace{10111010}_{0xBA} \underbrace{10111110}_{0xBE} \underbrace{11111111}_{0xFF} \underbrace{11010110}_{0xD6} \underbrace{11100101}_{0xE5} \underbrace{10101010}_{0xAA} \underbrace{01010101}_{0x55}$					
#	Binary	Hex	Dec	Sign	Gloss
0	10110111	0xB7	183	AN	sky / heaven god
1	10111010	0xBA	186	KI	earth
2	10111110	0xBE	190	EN.KI	lord of earth / waters
3	11111111	0xFF	255	DĠIR	divine determinative
4	11010110	0xD6	214	SI.SI	dragon seed
5	11100101	0xE5	229	E ₂	house / temple
6	10101010	0xAA	170	ZU	knowledge
7	01010101	0x55	85	UR	dog / watchman

Invariants. The declared base-10 checksum (the *EE/E7* viral-junction signature) is

$$\text{chk}(\mathbf{b}) = 1621, \quad (\text{raw byte sum } \sum_i b_i = 1512). \quad (8)$$

The binary seed $101_2 = 5$ is the Pentagon Seed / Fibonacci growth kernel.

Symbolic Interpretation. The eight signs read as a Sumerian sentence-glyph: heaven (*AN*) and earth (*KI*) under the lord *EN.KI*, marked divine (*DIĜIR*), carrying the dragon seed (*SI.SI*) into the temple-house (*E₂*) of knowledge (*ZU*), guarded (*UR*).

6 Machine-checked Fragment (Lean 4)

The discrete/algebraic content above is formalized and machine-checked in the companion file `cosmovirus.lean` (Lean 4, core only). Key signatures:

Lean declaration	Corresponds to
<code>phi_pow : Nat → Float</code>	Operator 2.1
<code>phi_mod (n m : Nat) : Nat</code>	$\sigma(n) = \lfloor \varphi^n \rfloor \bmod m$
<code>diag_apply (a b c : Float) : Float</code>	Operator 2.2
<code>diag_null_sum : diag_apply_ℤ a a a = 0</code>	kernel property
<code>inductive CosmoLayer</code>	objects of §4
<code>scl_projection / triality_rotation</code>	Operators 2.1 , 2.3
<code>infection / ouroboros</code>	HPV16, Operator 2.4
<code>psi_equation : CosmoLayer → CosmoLayer</code>	Eq. (6)
<code>ouroboros_fixed_point</code>	Prop. (loop closure)

Mathematical content is asserted formally; Symbolic Interpretation notes are non-mathematical readings and carry no formal force.